Inference Theory for Statement Calculus

- The main function of logic is to provide rules for inference.
- The rules for inference are criteria for determining the validity of an argument.
- These rules are stated in terms of premises(hypothesis or assumptions) and conclusions.
- If a conclusion is derived from a set of premises by using the accepted rules of reasoning, then such a process is called deduction or formal proof and argument or conclusion is called valid argument or valid conclusion.

Inference Theory

An argument is a sequence of propositions $H_1, H_2, \ldots, H_n$ called premises (or hypotheses) followed by a proposition C called conclusion. An argument is usually written:

$$\begin{array}{c} H_1 \\ H_2 \\ \vdots \\ H_n \\ \text{implies} \quad C \end{array}$$

or

$$H_1 \wedge H_2 \wedge \ldots \wedge H_n \Rightarrow C$$

The argument is valid if C is true whenever $H_1, H_2, \ldots, H_n$ are true; otherwise it is invalid.

Example: $H_1 : p$ and $H_2 : p \to q$ then $C : q$ (Modus Ponens)

p	q	$p \to q$	$p \wedge (p \to q)$	$(p \wedge (p \to q)) \to q$
T	T	T	T	T
T	F	F	F	T
F	T	T	F	T
F	F	T	F	T

Notice that $(p \wedge (p \to q)) \to q$ is a tautology. Therefore it is valid.

Thus $p,\ p \to q \Rightarrow q$.

Example: $H_1 : q$ and $H_2 : p \to q$ then $C : p$

p	q	$p \to q$	$q \land (p \to q)$	$(q \land (p \to q)) \to p$
T	T	T	T	T
T	F	F	F	T
F	T	T	T	F
F	F	T	F	T

Notice that $q \land (p \to q) \to p$ is a contigency. Therefore it is invalid.

Implication Table:

S.No	Formula	Name
1	$p \wedge q \Rightarrow p$	simplification
	$p \wedge q \Rightarrow q$	
2	$p \Rightarrow p \vee q$	addition
	$q \Rightarrow p \vee q$	
3	$p,\ q \Rightarrow p \wedge q$	
4	$p,\ p \rightarrow q \Rightarrow q$	modus ponens
5	$\neg p,\ p \vee q \Rightarrow q$	disjunctive syllogism
6	$\neg q,\ p \rightarrow q \Rightarrow \neg p$	modus tollens
7	$p \rightarrow q\, , q \rightarrow r \Rightarrow p \rightarrow r$	

Rules of inference:

Rule P: A premises can be introduced at any step of derivation.

Rule T: A formula can be introduced provided it is Tautologically implied by previously introduced formulas in the derivation.

Rule CP: If the conclusion is of the form $r \to s$ then we include r as an additional premises and derive s.

Indirect method: We use negation of the conclusion as an additional premise and try to arrive a contradiction.

Inconsistent: A set of premises are inconsistent provided their conjunction implies a contradiction.

Example: Show that $\neg q$ and $p \rightarrow q$ implies $\neg p$.

Solution: (A formal proof is as follows):

Step 1.	$p \rightarrow q$	Rule P
Step 2.	$\neg q \rightarrow \neg p$	Rule T
Step 3.	$\neg q$	Rule P
Step 4.	$\neg p$	Combined $\{2, 3\}$ and apply Modus Ponens

Example: Show that r is a valid inference from the premises $p \to q$, $q \to r$ and p.

Solution:

Step	Premises	Rule
1	p	Rule P
2	$p \to q$	Rule P
3	q	Rule T, {1,2} Modus Ponens
4	$q \to r$	Rule P
5	r	Rule T {3,4} Modus Ponens

Example: Show that $s \vee r$ is tautologically implied by $p \vee q$, $p \to r$ and $q \to s$.

Solution:

Step	Premises	Rule
1	$p \vee q$	Rule P
2	$\neg p \to q$	Rule T
3	$q \to s$	Rule P
4	$\neg p \to s$	Rule T,{2,3} Hypothetical Syllogism
5	$\neg s \to p$	Rule T , from 4,Contrapositive
6	$p \to r$	Rule P
7	$\neg s \to r$	Rule T,{5,6} Hypothetical Syllogism
8	$s \vee r$	Rule T

Example: Show that $R \wedge (P \vee Q)$ is a valid conclusion from the premises $P \vee Q, Q \rightarrow R, P \rightarrow M, \neg M$

Solution:

(1) $P \rightarrow M$	Rule P
(2) $\neg M$	Rule P
(3) $\neg P$	Rule T, (1), (2), and $\neg Q, P \rightarrow Q \Rightarrow \neg P$
(4) $P \vee Q$	Rule P
(5) Q	Rule T, (3), (4) and $\neg P, P \vee Q \Rightarrow Q$
(6) $Q \rightarrow R$	Rule P
(7) R	Rule T, (5), (6) and $P, R \rightarrow Q \Rightarrow Q$
(8) $R \wedge (P \vee Q)$	Rule T, (4), (7) and $P, Q \Rightarrow P \wedge Q$

Example: Show that $P \to Q, Q \to \rceil R, R, P \vee (J \wedge S) \Rightarrow J \wedge S$.

(1) $P \to Q$	Rule P
(2) $Q \to \rceil R$	Rule P
(3) $P \to \rceil R$	$\because P \to Q, Q \to R \Rightarrow P \to R$
(4) $R \to \rceil P$	$\because P \to Q \Leftrightarrow \rceil Q \Rightarrow \rceil P$
(5) R	Rule P
(6) $\rceil P$	$\because P, P \to Q \Rightarrow Q$
(7) $P \vee (J \wedge S)$	Rule P
(8) $J \wedge S$	$\because \rceil P, P \wedge Q \Rightarrow Q$.

Example: Consider the following statements: 'I take the bus or I walk. If I walk I get tired. I do not get tired. Therefore I take the bus.'

Solution:

We can formalize this by calling B= I take the bus, W= I walk and J= I get tired. The premises are $B \vee W$, $W \to J$ and $\neg J$, and the conclusion is B. The argument can be described in the following steps:

step	statement	reason
1	$W \to J$	P
2	$\neg J$	P
3	$\neg W$	$\{1, 2\}$, Modus Tollens
4	$B \vee W$	P
5	B	$\{3, 4\}$, Disjunctive Syllogism

Example:

Show that the premises "It is not sunny this afternoon and it is colder than yesterday," "We will go swimming only if it is sunny," "If we do not go swimming, then we will take a canoe trip," and "If we take a canoe trip, then we will be home by sunset" lead to the conclusion "We will be home by sunset."

Solution:

p: It is sunny this afternoon

q: It is colder than yesterday

r : we go swimming

s: We will take a canoe trip

t: We will be home by sunset (the conclusion)

1. $\neg p \wedge q$
2. $r \rightarrow p$
3. $\neg r \rightarrow s$
4. $s \rightarrow t$
5. t

We construct an argument to show that our premises lead to the desired conclusion as follows.

Step	Reason
1. $\neg p \wedge q$	Premise
2. $\neg p$	Simplification using (1)
3. $r \rightarrow p$	Premise
4. $\neg r$	Modus tollens using (2) and (3)
5. $\neg r \rightarrow s$	Premise
6. s	Modus ponens using (4) and (5)
7. $s \rightarrow t$	Premise
8. t	Modus ponens using (6) and (7)

Example:

"If there was a ball game, then travelling was difficult. If they arrived on time, then travelling was not difficult. They arrived on time. Therefore, there was no ball game". Show that these statements constitute a valid argument.

Solution:

Let P : There was a ball game

Q : Travelling was difficult

R : They arrived on time.

The premises are:

$P \to Q, R \to \rceil Q, R$ and the conclusion is $\rceil P$

1.	$P \to Q$	Rule P
2.	$R \to \rceil Q$	Rule P
3.	$Q \to \rceil R$	Rule T, (2), $P \to Q \Leftrightarrow \rceil Q \to \rceil P$
4.	$P \to \rceil R$	Rule T, (1), (3), $P \to Q, Q \to R \Rightarrow P \to R$
5.	$R \to \rceil P$	Rule T, (4), $P \to Q \Leftrightarrow \rceil Q \to \rceil P$
6.	R	Rule P
7.	$\rceil P$	Rule T, (5), (6), $P, P \to Q \Rightarrow Q$

Hence the given statements constitute a valid argument.

II. Problems:

(i) Show that the following argument is valid.
My father praises me only if i can be proud of myself. Either I do well in sports
or I cannot be proud of myself. If study hard, then I cannot do well in sports.
Therefore, if father praises me, then I do not study well.

Let us introduce a third inference rule known as rule *CP* or rule of conditional proof.

Rule CP: If the conclusion is of the form $r \to s$ then we include r as an additional premises and derive s.

Example: **Show that $R \to S$ can be derived from the premises $\mathrm{P} \to (Q \to S)$, $\sim R \vee P$ and Q.**

Instead of deriving $R \to S$, we shall include R as an additional premise and arrive at S. That is, to prove 'S' as a conclusion.

1.	$\rceil R \vee P$	Rule P
2.	$R \to P$	Rule T, (1); $P \to Q \Leftrightarrow \rceil P \vee Q$
3.	R	Rule P (Assumed Premise)
4.	P	Rule T, (2), (3); $P, P \to Q \Rightarrow Q$
5.	$P \to (Q \to S)$	Rule P
6.	$Q \to S$	Rule T, (4), (5), $P, P \to Q \Rightarrow Q$
7.	Q	Rule P
8.	S	Rule T, (6), (7), $P, P \to Q \Rightarrow Q$
9.	$R \to S$	Rule CP

Alternate Method [Using Rule CP]:

Solution:

Include P as an additional premise and arrive at S.

1.	$\neg P \vee Q$	Rule P
2.	$P \to Q$	Rule T, (1), $P \to Q \Leftrightarrow \neg P \vee Q$
3.	P	Rule P (Assumed Premise)
4.	Q	Rule T, (2), (3), $P, P \to Q \Rightarrow Q$
5.	$\neg Q \vee R$	Rule P
6.	$Q \to R$	Rule T, (5), $P \to Q \Rightarrow \neg P \vee Q$
7.	R	Rule T, (4), (6), $P, P \to Q \Rightarrow Q$
8.	$R \to S$	Rule P
9.	S	Rule T, (7), (8), $P, P \to Q \Rightarrow Q$
10.	$P \to S$	Rule CP.

Example:

Derive $P \to (Q \to S)$, using rule CP from the premises $P \to (Q \to R), Q \to (R \to S)$

Solution:

We include P as an additional premise.

1.	$P \to (Q \to R)$	Rule P
2.	P	Rule P
3.	$Q \to R$	Rule T, (1), (2), $P, P \to Q \Rightarrow Q$
4.	$Q \to (R \to S)$	Rule P
5.	$\rceil Q \vee (R \to S)$	Rule T, (4), $P \to Q \Leftrightarrow \rceil P \vee Q$
6.	$\rceil Q \vee R$	Rule T, (3), $P \to Q \Leftrightarrow \rceil P \vee Q$
7.	$\rceil Q \vee (R \wedge (R \to S))$	Rule T, (5), (6) Distributive law
8.	$\rceil Q \vee S$	Rule T, (7), $P, P \to Q \Rightarrow Q$
9.	$Q \to S$	Rule T, (8), $P \to Q \Leftrightarrow \rceil P \vee Q$
10.	$P \to (Q \to S)$	Rule CP.

INDIRECT METHOD OF PROOF:

The technique of indirect method of proof run as follows:

1. Introduce the negation of the desired conclusion as a new premise.
2. Derive a contradiction from the set of premises including the new premise.

Example: Prove by indirect method that $p \to q$, $p \vee r, \neg q$ implies r.
Solution: The desired result is r. Include $\neg r$ as a new premise.

Step	Premises	Rule
1	$p \vee r$	Rule P
2	$\neg r \to p$	Rule T
3	$\neg r$	Rule P,(additional premise)
4	p	Rule T, {2,3) Modus Ponens
5	$p \to q$	Rule P
6	q	Rule T, {4,5} Modus Ponens
7	$\neg q$	Rule P
8	$q \wedge \neg q$	Rule T, {6,7}

The new premise together with the given premises, leads to a contradiction.
Thus $p \to q$, $p \vee r, \neg q$ implies r.

Example: Show by indirect method that $\rceil(P \wedge Q)$ follows from $\rceil P \wedge \rceil Q$.

Solution:

Include $\rceil(\rceil(P \wedge Q))$ as an additional premise.

(i.e) $\rceil\rceil(P \wedge Q) \Leftrightarrow P \wedge Q$ as an additional premise.

$(\because \rceil\rceil P \Leftrightarrow P)$

1.	$P \wedge Q$	Rule P, Additional premise
2.	P	Rule T, (1), $P \wedge Q \Rightarrow P$
3.	$\rceil P \wedge \rceil Q$	Rule P
4.	$\rceil P$	Rule T, (3), $P \wedge Q \Rightarrow P$
5.	$P \wedge \rceil P \Leftrightarrow \mathbf{F}$	Rule T, (2), (4), $P, Q \Rightarrow P \wedge Q$

Thus the negation of conclusion as an additional premise leads to a contradiction.

Example: Using indirect method prove that

$$(R \to \neg Q), R \vee S, S \to \neg Q, P \to Q \Rightarrow \neg P$$

Solution: Introduce the negation of the desired conclusion, i.e, P as an additional premise

1.	P	Rule P, Additional premise
2.	$P \to Q$	Rule P
3.	Q	Rule T, (1), (2), $P, P \to Q \Rightarrow Q$
4.	$S \to \rceil Q$	Rule P
5.	$Q \to \rceil S$	Rule T, (4), $P \to Q \Leftrightarrow \rceil Q \to \rceil P$
6.	$\rceil S$	Rule T, (3), (5), $P, P \to Q \Rightarrow Q$
7.	$R \vee S$	Rule P
8.	$\rceil R \to S$	Rule T, (7), $P \to Q \Leftrightarrow \rceil P \vee Q$
9.	$\rceil S \to R$	Rule T, (8), $P \to Q \Leftrightarrow \rceil Q \to \rceil P$
10.	R	Rule T, (6), (9), $P, P \to Q \Rightarrow Q$
11.	$R \to \rceil Q$	Rule P
12.	$\rceil Q$	Rule T, (10), (11), $P, P \to Q \Rightarrow Q$
13.	$Q \wedge \rceil Q \Leftrightarrow \mathbf{F}$	Rule T, (3), (12), $P, Q \Rightarrow P \wedge Q$

Inconsistent: A set of premises are inconsistent provided their conjunction implies a contradiction.

i.e. If $H_1 \wedge H_2 \wedge \ldots \wedge H_n \Rightarrow \mathbf{F}$

Example: Prove that $p \to q$, $p \to r$, $q \to \neg r$ and p are inconsistent.
Solution: The desired result is false.

Step	Premises	Rule
1	p	Rule P
2	$p \to q$	Rule P
3	q	Rule T, {1,2} Modus Ponens
4	$q \to \neg r$	Rule P
5	$\neg r$	Rule T, {3,4} Modus Ponens
6	$p \to r$	Rule P
7	$\neg p$	Rule T, {5,6) Modus Tollens
8	F	Rule T, $\neg p \land p \Leftrightarrow F$

(i) Show that the following set of premises is inconsistent.

1. If Jack misses many classes through illness, then he fails high school.
2. If Jack fails high school, then he is uneducated.
3. If Jack reads a lot of books, then he is not uneducated.
4. Jack misses many classes through illness and reads a lot of books.

Let us consider,
E : Jack misses many classes through illness
S : Jack fails high school
A : Jack reads a lot of books
H : Jack is uneducated.
The premises are,

- $E \rightarrow S$
- $S \rightarrow H$
- $A \rightarrow \neg H$ and
- $E \wedge A$

Step	Premises	Rule
1	$E \wedge A$	Rule P
2	E	Rule T
3	A	Rule T
4	$E \rightarrow S$	Rule P
5	S	Rule T, {2,4} Modus Ponens
6	$S \rightarrow H$	Rule P
7	H	Rule T, {5,6} Modus Ponens
8	$A \rightarrow \neg H$	Rule P
9	$\neg H$	Rule T, {3,8}, Modus Ponens
10	$H \wedge \neg H$	Rule T, {7,9}

Example: **Test the validity of the following argument: If I study, then I will pass in the examination. If I watch TV, then I will not study. I failed in the examination. Therefore I watched TV.**

Solution:

Let P : I will pass in the examination

S : I will study

W : I watch TV.

The given premises are:

$S \to P, W \to \neg S, \neg P$ and given conclusion is W.

1.	$S \to P$	Rule P
2.	$\neg P$	Rule P
3.	$\neg S$	Rule T, (1), (2), $\neg Q, P \to Q \Rightarrow \neg P$
4.	$W \to \neg S$	Rule P
5.	$\neg S$	Rule T, (3), (4), $Q, P \to Q \Rightarrow Q$

Since we did not arrive at conclusion W, as the valid conclusion is $\neg S$. So the given argument is not valid.

Example: Show that the following set of premises are inconsistent.

If war is near, then the army would be mobilized.

If the army has mobilized then labour costs are high.

However the war is near and yet labour costs are not high.

Example: Show that the following set of premises is inconsistent.

If the contract is valid, then John is liable for penalty. If John is liable for penalty,
he will go bankrupt. If the bank will loan him money, he will not go bankrupt.
As a matter of fact, the contract is valid and the bank will loan him money.

Solution:

Let	C	:	The contract is valid
	P	:	John is liable for penalty
	L	:	Bank will loan him money
	B	:	He will go bankrupt.

The given premises are:

$$C \to P, P \to B, L \to \neg B, C \wedge L$$

To prove,

$$C \to P, P \to B, L \to \neg B, C \wedge L \Rightarrow \mathbf{F}$$

1.	$C \to P$	Rule P
2.	$P \to B$	Rule P
3.	$C \to B$	Rule T, (1), (2), $P \to Q, Q \to R \Rightarrow P \to R$
4.	$C \wedge L$	Rule P
5.	C	Rule T, (4) $P \wedge Q \Rightarrow P$
6.	L	Rule T, (4), $P \wedge Q \Rightarrow Q$
7.	B	Rule T, (3), (5), $P, P \to Q \Rightarrow Q$
8.	$L \to \rceil B$	Rule P
9.	$B \to \rceil L$	Rule T, (8), $P \to Q \Leftrightarrow \rceil Q \to \rceil P$
10.	$\rceil L$	Rule T, (7), (9), $P, P \to Q \Rightarrow Q$
11.	$L \wedge \rceil L \Leftrightarrow \mathbf{F}$	Rule T, (6), (10), $P, Q \Rightarrow P \wedge Q$

Since the given set of premises leads to a contradiction, it is inconsistent.